

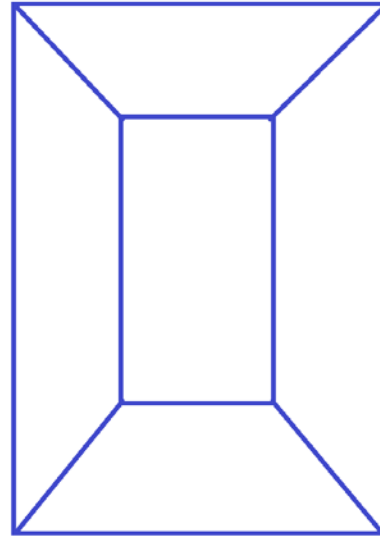
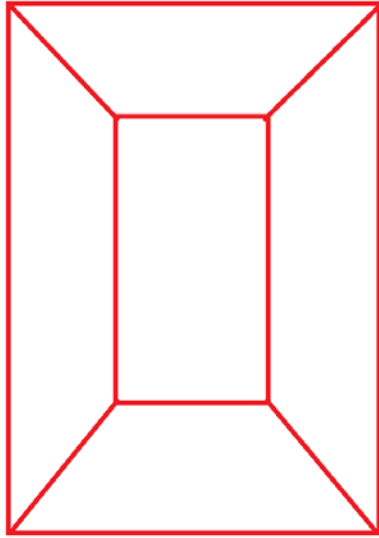
Hello, HPM Group

We have endeavoured above to employ only symmetrical figures, such as should not merely be an aid to the sense of sight, but should also be to some extent elegant in themselves. But for merely theoretical purposes the rule of formation would be very simple. It would merely be to begin by drawing any closed figure, and then proceed to draw others, subject to the one condition that each is to intersect once and once only all the existing subdivisions produced by those which had gone before. Proceeding thus we should naturally select circles as the simplest figures, so long as they would answer our purpose; that would be, up to three terms inclusive. The two successive modifications, aiming always at simplicity of figure, would then be naturally such as the following (the fifth figure is marked, for clearness, by a dotted line):—



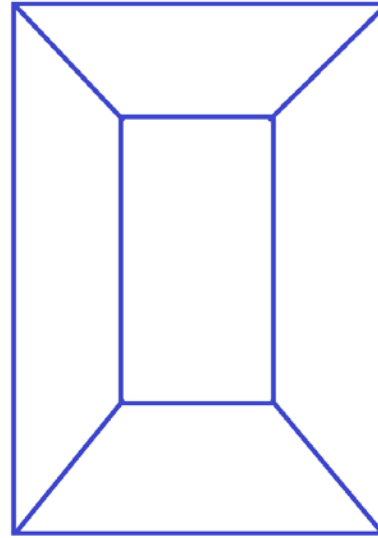
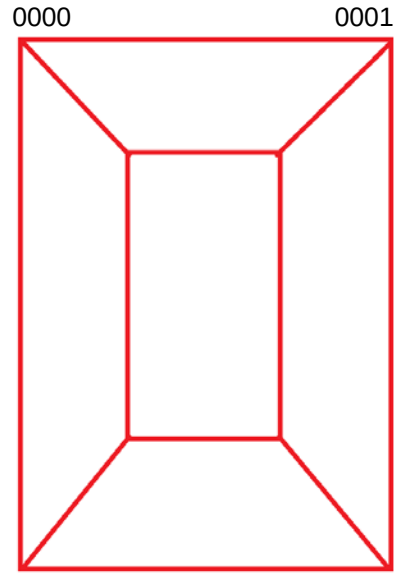
Symmetry does come in naturally if we look into $\{0,1\}^4$

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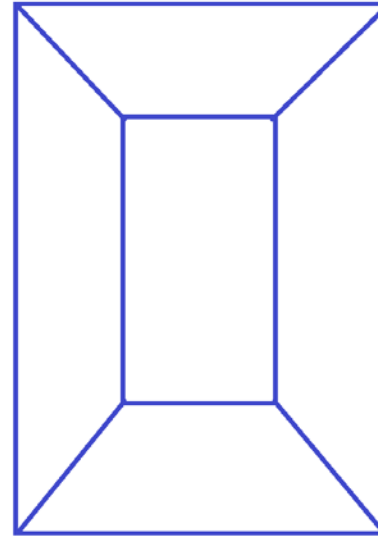
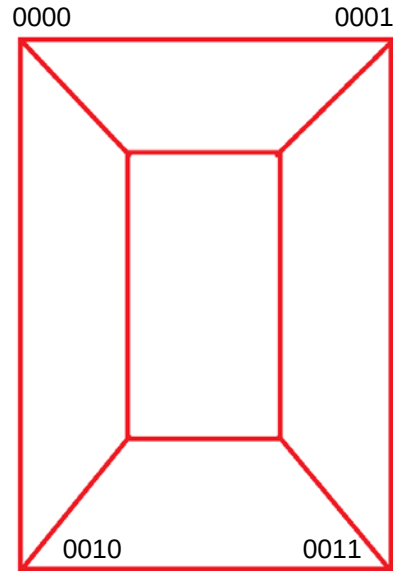
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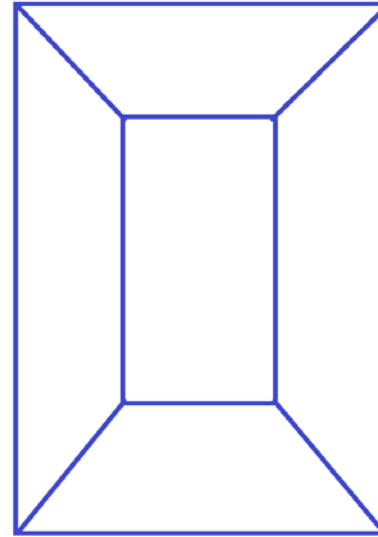
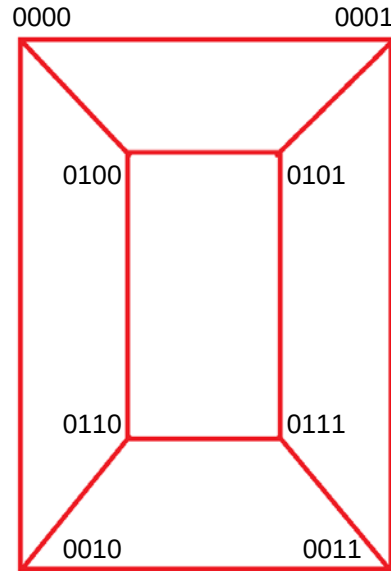
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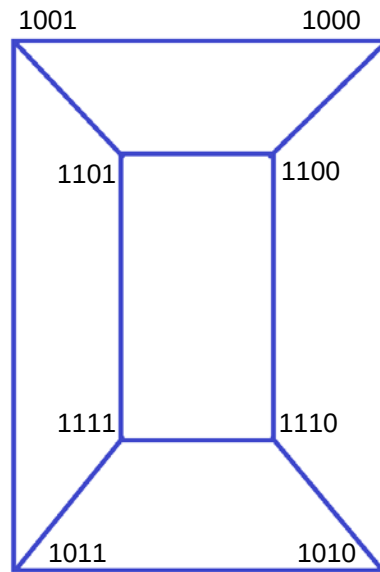
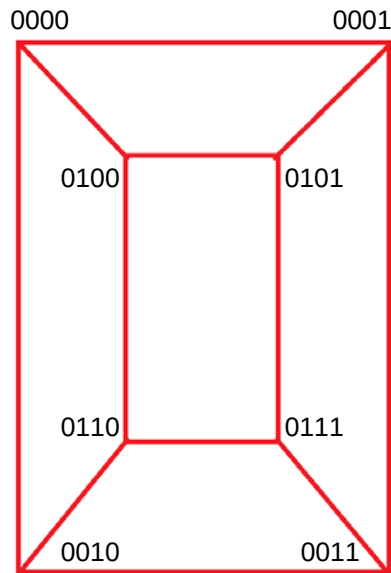
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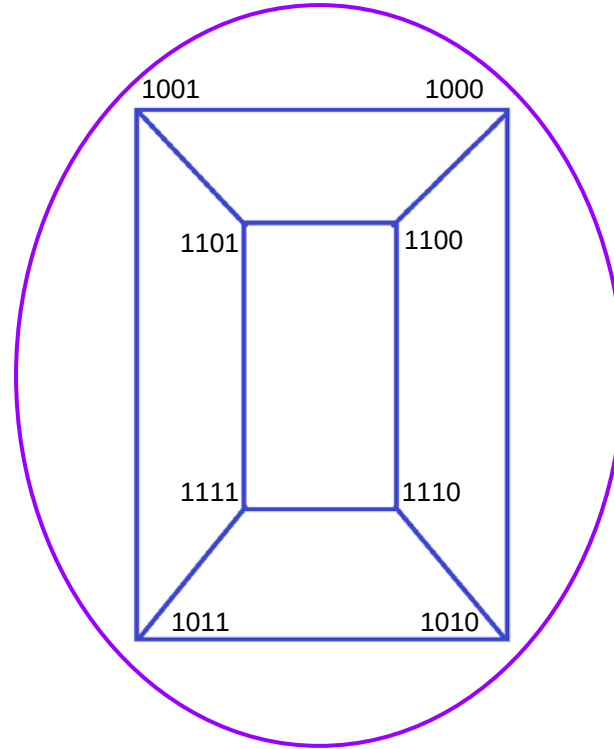
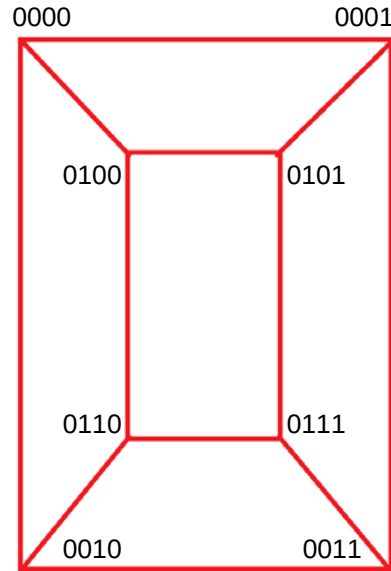
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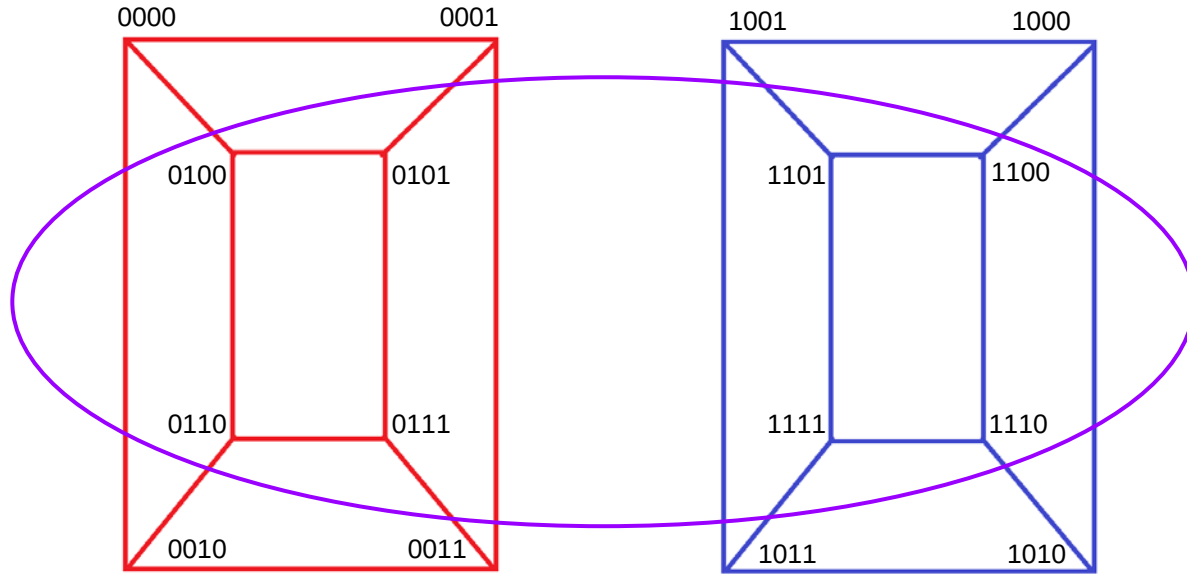
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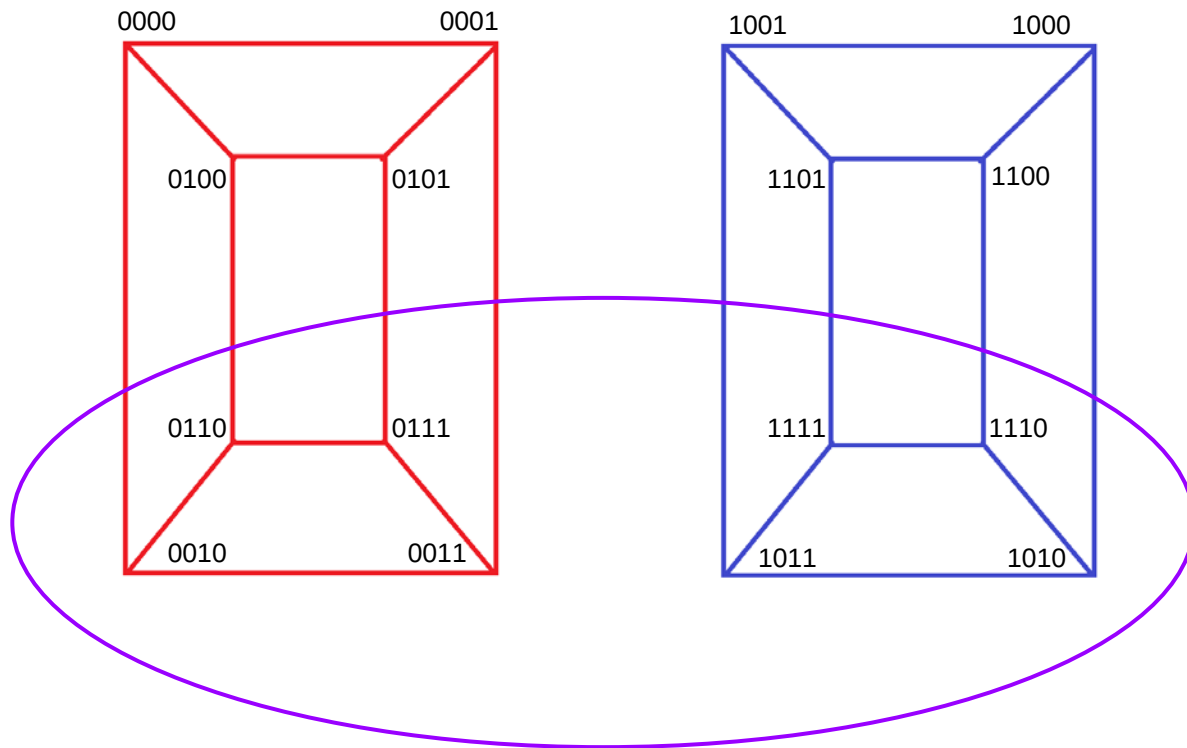
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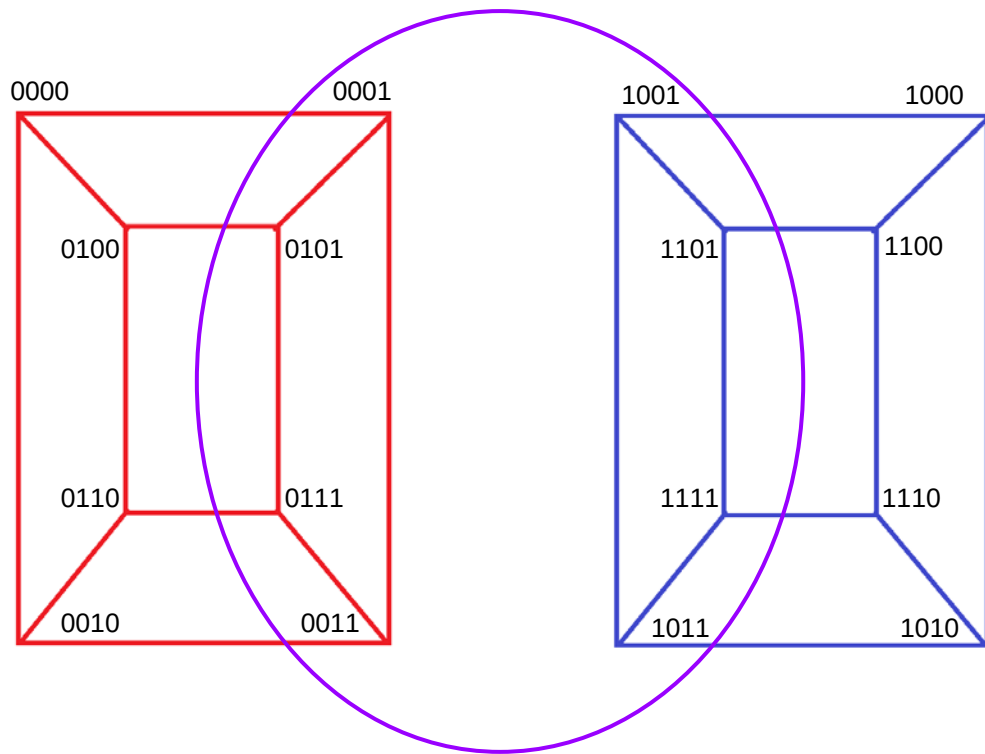
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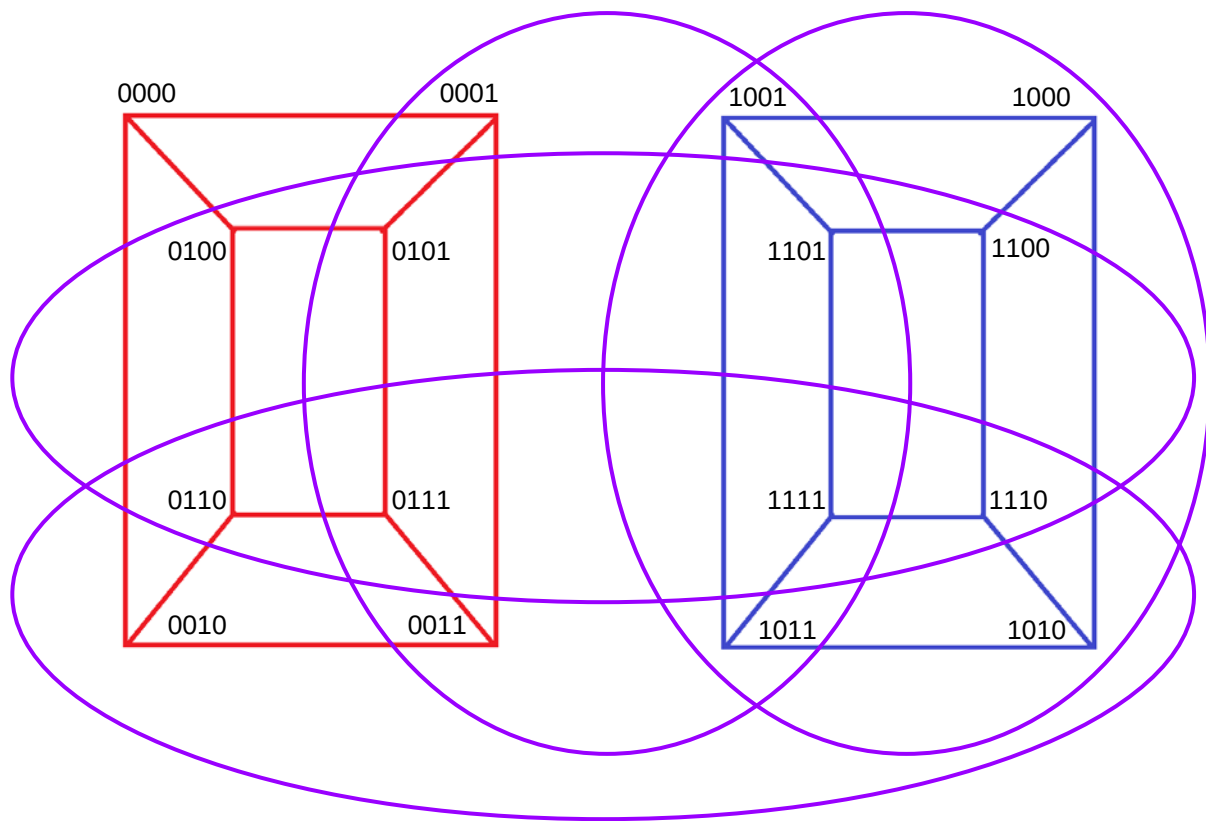
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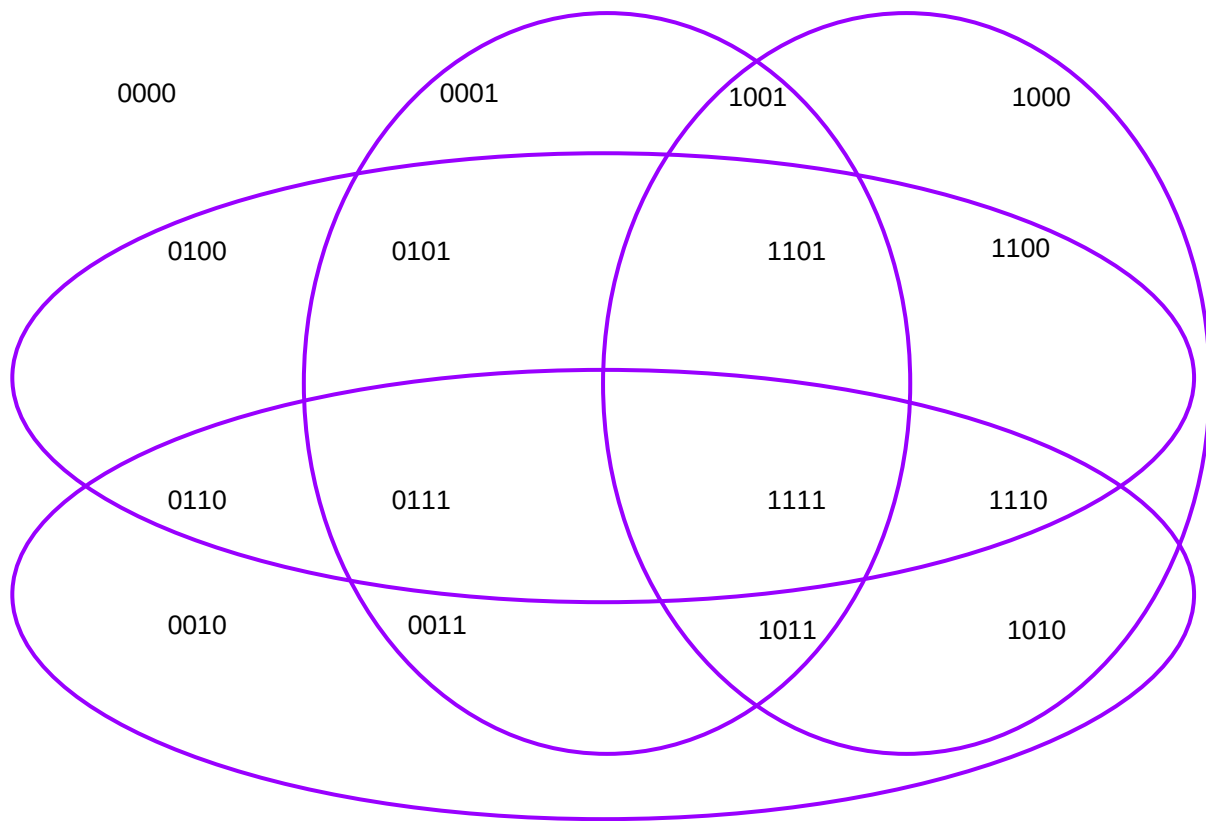
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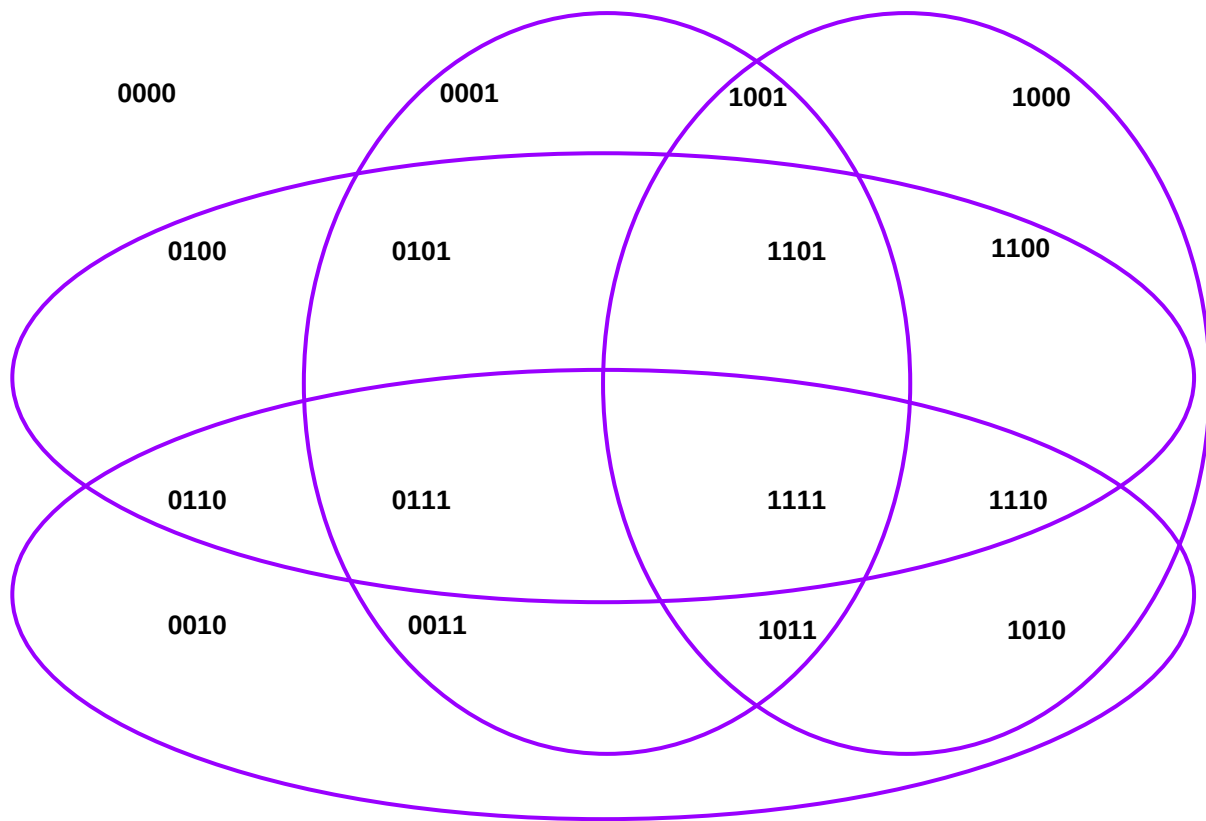
Just a puzzle. YK.

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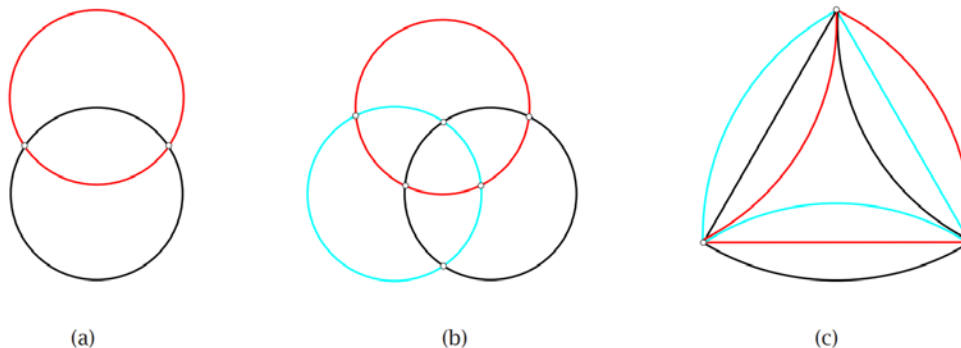


Just a puzzle. YK.

So, we are given more time...

Many people are surprised to learn that Venn diagrams can be drawn to represent all of the intersections of more than three sets. This surprise is perfectly understandable since small Venn diagrams are often drawn with circles, and it is impossible to draw a Venn diagram *with circles* that will represent all the possible intersections of four (or more) sets. This is a simple consequence of the fact that circles can finitely intersect in at most two points and Euler's relation $F - E + V = 2$ for the number of faces, edges, and vertices in a plane graph. However, there is no reason to restrict the curves of a Venn diagram to be circles; in modern definitions a Venn diagram is a collection of simple closed Jordan curves. This collection must have the property that the curves intersect in only finitely many points and the property that the intersection of the interiors of any of the 2^n sub-collections of the curves is a nonempty connected region.

If a Venn diagram consists of n curves then we call it an n -Venn diagram. The *rank* of a region is the number of curves that contain it. In any n -Venn diagram there are exactly $\binom{n}{r}$ regions of rank r . Figure 1 shows a 2-Venn and two distinct 3-Venn diagrams. Note that the diagram in Figure 1(c) has three points where all three curves intersect. The regions in the diagrams of Figure 2 are colored according to rank.



Symmetric n -Venn diagrams for $n = 2, 3$: (a) $n = 2$, (b) $n = 3$ simple, (c) $n = 3$ non-simple.

So, we are given more time...

The traditional three-circle Venn diagram has an appealing 3-fold rotational symmetry, and it is natural to ask whether there are n -Venn diagrams with an n -fold rotational symmetry for $n > 3$. Grünbaum [6] found a symmetric 5-Venn diagram made from ellipses. Henderson [10] noted the following necessary condition: if an n -Venn diagram has an n -fold rotational symmetry, then n is prime. The reason is as follows:

In any symmetric n -Venn diagram the fixed point of the rotations, the center of the diagram, must lie in the unique region of rank n . The unbounded outer region has rank 0. Regions of rank $0 < r < n$ must be distributed symmetrically and thus their number, $\binom{n}{r}$, must be divisible by n . This property holds exactly when n is prime.

Why? Recall that $\binom{n}{r} = n(n-1) \cdots (n-r+1)/r!$. If n is prime and $0 < r < n$, then note that n occurs once in the right-hand side and all other numbers are less than n . On the other hand, if p is a nontrivial divisor of n , then the binomial coefficient with $r = p$ is the product of two integers $\binom{n}{p} = \frac{n}{p} \cdot m$ where $m = (n-1) \cdots (n-p+1)/(p-1)!$, but clearly p cannot divide m , and thus n does not divide $\binom{n}{p}$.

The elegant necessary condition of Henderson was long suspected to be sufficient, but it took some 40 years before it was proven to be sufficient by Griggs, Killian, and Savage [5]. In the intervening years, symmetric diagrams were discovered for n equal to 5, 7, and 11. Some of these diagrams are shown in Figure 2. The first symmetric 7-Venn diagrams were discovered independently by Grünbaum [7] and Edwards [3] (Fig. 2(b)); the first symmetric 11-Venn diagram was discovered by Hamburger [8].

About the Speaker



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